

A Strictly Positive-Field Counterexample to the Boundary–Bergman Equality Question

Automated mathematics run `fa_banach.001`, lane 10

13 August 2026

Status. Candidate full counterexample, likely valid, pending human review. A smooth polynomial magnetic field satisfying $B > 0$ makes the first inequality of Fournais–Frank–Goffeng–Kachmar–Sundqvist Proposition 5.2 strict.

1 Source question

Fournais, Frank, Goffeng, Kachmar, and Sundqvist study magnetic Pauli operators on planar domains in [1]. Their Proposition 5.2 gives, in particular,

$$N(\mathcal{H}_{\mathbf{A},g}^{\mathcal{O}}, 0) \geq \dim(E_V \cap \gamma_0(\mathcal{B}(\Omega))), \quad V = g - A_{\tau}. \quad (1)$$

Remark 5.3 on PDF page 18 asks whether the inequalities in that proposition are equalities when $B \geq 0$. The latest source version renumbers the same passage as Proposition 6.3 and Remark 6.4.

We disprove the proposed first equality with a field that is strictly positive on the closed disc. This suffices to answer the general question negatively.

2 Definitions and proof intuition

Let $\mathbb{D} = \{z \in \mathbb{C} : |z| < 1\}$ and impose Neumann data $g = 0$. Following the source, choose a real function ϕ vanishing on $\partial\mathbb{D}$ and put

$$\mathbf{A} = (-\partial_y \phi, \partial_x \phi), \quad B = \operatorname{curl} \mathbf{A} = \Delta \phi.$$

The source orients the boundary tangent counterclockwise (and the normal inward), so

$$A_{\tau} = \mathbf{A} \cdot \tau = \partial_r \phi|_{r=1}.$$

On holomorphic functions the compressed quadratic form is

$$q_{\mathbf{A},0}^{\mathcal{O}}(v) = \int_0^{2\pi} \left((-i\partial_{\theta} - A_{\tau}(\theta))v(e^{i\theta}) \right) \overline{v(e^{i\theta})} d\theta. \quad (2)$$

Proof intuition. We prescribe A_{τ} to have only Fourier frequencies 0 and ± 3 and realize it by a polynomial ϕ whose Laplacian remains positive. Formula (2) then splits into three blocks according to Fourier index modulo three. Two blocks contain the plainly negative modes $1, z$, while a 2×2 determinant forces a negative direction in the third block. The comparison space in (1) is nevertheless exactly two-dimensional: the explicit boundary-Dirac eigenfunctions admit a Wiener–Hopf factorization, reducing holomorphicity to the preimage of $\operatorname{span}\{1, z\}$ under an invertible coanalytic Toeplitz operator.

for $\partial_{\bar{z}}$. Therefore F is in the form domain of the operator $\mathcal{H}_{\mathbf{A},g}^{\mathcal{O}}$. The proof of the inequality $N(\mathcal{H}_{\mathbf{A},g}^{\mathcal{O}}, 0) \geq \dim(E_V \cap \gamma_0(\mathcal{B}(\Omega)))$ now follows by Glazman's lemma.

In a similar fashion, the inequality $N(\mathcal{H}_{\mathbf{A},g}^{\mathcal{O}}, 0_+) \geq \dim(F_{V^c} \cap \gamma_0(\bar{\mathcal{B}}(\Omega)))$ follows. $\square$

Remark 5.3. It is unclear to us when the opposite inequalities in Proposition 5.2 can be expected to hold true. For instance, a natural question is whether we have equalities in Proposition 5.2 when $B \geq 0$. Let us point at the technical difficulty. Let $N = N(\mathcal{H}_{\mathbf{A},g}^{\mathcal{O}}, 0)$ and consider an orthonormal basis $\{u_1, \dots, u_N\}$ of the eigenspace $M = \text{Im}(\mathbf{1}_{(-\infty, 0)}(\mathcal{H}_{\mathbf{A},g}^{\mathcal{O}}))$, corresponding to the negative eigenvalues of $\mathcal{H}_{\mathbf{A},g}^{\mathcal{O}}$. Then $\dim(\gamma_0(M)) = N$ but we do not necessarily have the inclusion $\gamma_0(M) \subset E_V \cap \gamma_0(\mathcal{B}(\Omega))$. Even after projecting again on E_V , we do not know that the set $\{\Pi_V \gamma_0(u_1), \dots, \Pi_V \gamma_0(u_N)\}$ is linearly independent, or even a subset of $\gamma_0(\mathcal{B}(\Omega))$.

When B is allowed to change sign, the inequalities in Proposition 5.2 can be strict. Indeed, with $\Omega = D(0, 1)$ and $B(x) = -4x_2$ the first inequality in Proposition 5.2 is strict. In this setup, $\phi(x) = \frac{x_2}{2}(1 - x_1^2 - x_2^2)$, $\mathbf{A}(x) = (-\frac{1}{2}(1 - x_1^2 - 3x_2^2), -x_1x_2)$, $A_\tau(s) = -\sin(s)$ and $g = 0$ (so $V(s) = -A_\tau(s)$), and B has zero flux. Let $v(z) = e^{-iz/2}$. Then v is holomorphic in the interior of Ω and

$$\begin{aligned} q_{B,0}^{\mathcal{O}}(e^\phi v) &= \int_0^{2\pi} [(-i\partial_s + \sin(s))v(e^{is})] \overline{v(e^{is})} ds \\ &= -\frac{1}{2} \int_0^{2\pi} e^{-is} e^{\cos s} ds \\ &= -\pi I_1(1) < 0, \end{aligned}$$

where I_1 denotes the modified Bessel function of the first kind. By Glazman's

Figure 1: Full-width source crop from arXiv:2307.16079v1, PDF page 18, containing Remark 5.3 and the exact equality question.

3 Counterexample theorem

Theorem 1. *Let*

$$a = \frac{399}{200}, \quad b = \frac{6}{25},$$

and define on the unit disc

$$\phi(r, \theta) = \frac{a}{2}(r^2 - 1) - b(r^3 - r^5) \cos(3\theta), \quad \mathbf{A} = (-\phi_y, \phi_x). \quad (3)$$

Then ϕ and $\mathbf{A}$ are smooth polynomials, $\phi|_{\partial\mathbb{D}} = 0$, and

$$B = \text{curl } \mathbf{A} \geq \frac{3}{20} > 0 \quad \text{on } \bar{\mathbb{D}}.$$

For Neumann data $g = 0$, the first inequality in Proposition 5.2 of [1] is strict:

$$N(\mathcal{H}_{\mathbf{A},0}^{\mathcal{O}}, 0) \geq 3 > 2 = \dim(E_{-A_\tau} \cap \gamma_0(\mathcal{B}(\mathbb{D}))). \quad (4)$$

In particular, nonnegativity of the magnetic field does not imply equality.

Proof. We divide the proof into the field construction, the negative-subspace calculation, and the exact comparison-space calculation.

1. The field is polynomial and strictly positive. Writing

$$P_3(x, y) = x^3 - 3xy^2 = r^3 \cos(3\theta),$$

formula (3) becomes

$$\phi(x, y) = \frac{a}{2}(x^2 + y^2 - 1) - b(1 - x^2 - y^2)P_3(x, y),$$

so it is a polynomial and vanishes on the circle. Direct differentiation gives

$$A_r(\theta) = \partial_r \phi(1, \theta) = a + 2b \cos(3\theta), \quad B(r, \theta) = \Delta \phi = 2a + 16br^3 \cos(3\theta). \quad (5)$$

Since $0 \leq r \leq 1$,

$$B(r, \theta) \geq 2a - 16b = \frac{399}{100} - \frac{96}{25} = \frac{3}{20} > 0.$$

2. The compressed form has at least three negative eigenvalues. For a holomorphic polynomial $v(z) = \sum_{n \geq 0} c_n z^n$, Fourier orthogonality and (5) give

$$\frac{1}{2\pi} q_{\mathbf{A},0}^{\mathcal{O}}(v) = \sum_{n \geq 0} (n - a) |c_n|^2 - b \sum_{n \geq 0} (c_{n+3} \overline{c_n} + c_n \overline{c_{n+3}}), \quad (6)$$

where coefficients outside the polynomial's support are zero. Thus the form is block diagonal according to n modulo three.

The directions 1 and z belong to two distinct residue blocks and have form values $-2\pi a$ and $2\pi(1 - a)$, both negative. In the third residue block, the matrix of $q_{\mathbf{A},0}^{\mathcal{O}}/(2\pi)$ on $\text{span}\{z^2, z^5\}$ is

$$M = \begin{pmatrix} 2 - a & -b \\ -b & 5 - a \end{pmatrix}.$$

Its trace is positive, whereas

$$\begin{aligned} \det M &= (2 - a)(5 - a) - b^2 \\ &= \frac{1}{200} \frac{601}{200} - \left(\frac{6}{25} \right)^2 = -\frac{1703}{40000} < 0. \end{aligned}$$

Hence M has one negative eigenvalue. Choose a corresponding negative vector $w \in \text{span}\{z^2, z^5\}$. The three vectors $1, z, w$ lie in distinct residue blocks, so they are mutually orthogonal for the quadratic form and their span is negative definite. The min-max principle (equivalently, Glazman's lemma) proves

$$N(\mathcal{H}_{\mathbf{A},0}^{\mathcal{O}}, 0) \geq 3.$$

3. The boundary comparison space has dimension two. Set $V = -A_r = -a - 2b \cos(3\theta)$. The explicit spectral formula for $\mathcal{D}_V = -i\partial_\theta + V$ in [1, Proposition 3.1] gives eigenvalues $m - a$, $m \in \mathbb{Z}$, and corresponding eigenfunctions, up to normalization,

$$f_m(z) = z^m E(z), \quad E(z) = \exp(\delta(z^3 - z^{-3})), \quad \delta = \frac{b}{3}, \text{quad} |z| = 1. \quad (7)$$

Indeed, the source formula uses $\int_0^\theta V = -a\theta - (2b/3) \sin(3\theta)$. Because $1 < a < 2$, the negative eigenvalues are precisely those with $m \leq 1$. Thus

$$E_V = E \overline{\text{span}\{z^m : m \leq 1\}}^{L^2(\partial \mathbb{D})}. \quad (8)$$

Let H^2 denote the usual Hardy space of nonnegative boundary Fourier modes. The source proves that $E_V \cap \gamma_0(\mathcal{B}(\mathbb{D})) \subset H^{1/2}(\partial\mathbb{D})$; hence every vector in the intersection is in H^2 . Conversely an H^2 boundary function is the trace of a holomorphic Hardy function and therefore of a Bergman function. It is consequently enough to calculate $E_V \cap H^2$.

Factor

$$E = E_+ E_-, \quad E_+(z) = \exp(\delta z^3), \quad E_-(z) = \exp(-\delta z^{-3}).$$

Suppose $f = Eh \in H^2$, where h has Fourier support in $\{m : m \leq 1\}$ as in (8), and put $G = E_+^{-1}f \in H^2$. Then

$$h = E_-^{-1}G = \bar{\alpha}G, \quad \alpha(z) = \exp(\delta z^3) = E_+(z) \text{quad}(|z| = 1).$$

Writing P_+ for the Hardy projection, the support condition on h is equivalent to

$$\mathsf{T}_{\bar{\alpha}}G := P_+(\bar{\alpha}G) \in \text{span}\{1, z\}. \quad (9)$$

The functions α and α^{-1} are bounded analytic functions on $\mathbb{D}$. Coanalytic Toeplitz operators multiply in the same order, and hence

$$\mathsf{T}_{\bar{\alpha}}\mathsf{T}_{\alpha^{-1}} = \mathsf{T}_{\overline{\alpha\alpha^{-1}}} = I, \quad \mathsf{T}_{\alpha^{-1}}\mathsf{T}_{\bar{\alpha}} = I.$$

Therefore $\mathsf{T}_{\bar{\alpha}}$ is invertible. The solution space of (9) is the inverse image of the two-dimensional space $\text{span}\{1, z\}$ and consequently has dimension exactly two. There is no regularity deficit: for $G = 1, z$, the vectors

$$f = E_+, zE_+$$

are smooth holomorphic traces, while $h = E_-^{-1}, zE_-^{-1}$ has Fourier support at most one. We conclude that

$$\dim(E_V \cap \gamma_0(\mathcal{B}(\mathbb{D}))) = 2.$$

Together with the preceding lower bound this proves (4). $\square$

4 A parameter family and robustness

The rational constants in Theorem 1 are only a convenient exact choice. The proof gives the following family.

Lemma 1. *Use (3) with real parameters satisfying*

$$1 < a < 2, \quad b \neq 0, \quad a \geq 8|b|, \quad (2-a)(5-a) < b^2.$$

Then $B \geq 0$, the comparison dimension is two, and the compressed form has at least three negative eigenvalues. If $a > 8|b|$, then $B > 0$ on $\overline{\mathbb{D}}$.

Proof. Equations (5)–(6) remain valid. The condition $a \geq 8|b|$ is exactly $2a - 16|b| \geq 0$. The determinant condition forces a negative direction in the z^2, z^5 block, while $1 < a < 2$ gives the same two negative low modes and the same cutoff $m \leq 1$ in the Dirac spectrum. The Toeplitz argument is unchanged. $\square$

For each sufficiently small fixed $b \neq 0$, taking $a < 2$ sufficiently close to two satisfies the determinant condition; the positivity condition is then also satisfied. The strict inequalities in Theorem 1 show that its example lies inside an open set of positive-field counterexamples.

5 Verification, scope, and novelty

The bundled script `code/exact_symbolic_check.py` verifies with exact symbolic and rational arithmetic the polynomial Laplacian, the boundary tangential potential, the lower bound $B \geq 3/20$, and the determinant $-1703/40000$. The Hardy/Toeplitz calculation is an analytic proof and is not inferred from a truncation.

Only the first equality proposed in Remark 5.3 is refuted. This is enough to answer the general question negatively. We do not claim a characterization for equality in the second, antiholomorphic inequality.

A bounded novelty search on 13 August 2026 covered the local run indexes and attempts, exact-title and exact-phrase web/arXiv queries, the arXiv id, the remark numbers, all five authors, and combinations of the core terms “nonnegative magnetic field”, “Bergman compression”, and “equality”. It found the source preprint and its 2025 journal publication, but no later claimed resolution or matching counterexample. The latest source TeX still contains the question. This supports moderate, not definitive, novelty confidence.

Human review should focus on the boundary orientation sign, the cutoff $m \leq 1$ in (8), and the equivalence (9). No unproved dependency is currently known.

References

- [1] S. Fournais, R. L. Frank, M. Goffeng, A. Kachmar, and M. Sundqvist, *Counting Negative Eigenvalues for the Magnetic Pauli Operator*, arXiv:2307.16079v1 (2023); Duke Math. J. **174** (2025), 313–353.